

FitBuddy – AI Fitness Plan Generator using Gemini Models

2. Prerequisites & Project Workflow

Gemini API Familiarity: [Google Generative AI Documentation](#)

HTML, CSS, and Jinja2 Template Skills: [W3Schools HTML/CSS/Jinja2 Tutorials](#)

Python Programming Proficiency: [Python Official Docs](#)

Version Control with Git: [Git Documentation](#)

SQLAlchemy and SQLite Basics: [SQLAlchemy Docs](#)

Environment Setup (pip, virtualenv, or conda): [Virtualenv Guide](#)

Uvicorn ASGI Server: [Uvicorn Docs](#)

Project Workflow:

Milestone 1. Model selection and Architecture

Activity 1.1. Research and Select the Appropriate Generative AI Model

Activity 1.2. Define the architecture of the application

Activity 1.3. Set up the development Environment

Milestone 2. Core functionalities Development

Activity 2.1. Develop the core functionalities

Activity 2.2. Implement the FastAPI Backend to Manage Routing and User Input Processing

Milestone 3. route.py Development

Activity 3.1. Writing the Main Application Logic in routes.py

Milestone 4. Frontend Development

Activity 4.1. Designing and Developing User Interface

Activity 4.2. Creating Dynamic Templates with FastAPI's Jinja2

Milestone 5. Deployment

Activity 5.1. Preparing the Application for Local Deployment

Activity 5.2. Testing and Verifying Local Deployment

Milestone 1. Model selection and Architecture

In this milestone, we focus on selecting the appropriate generative AI models from Google's Gemini family for the FitBuddy application. This includes evaluating the performance and capabilities of multiple Gemini models to ensure they align with the application's core functionalities—such as generating personalized workout plans, updating them based on feedback, and providing tailored nutrition or recovery advice.

Activity 1.1: Research and Select the Appropriate Generative AI Model

Understand the Project Requirements

FitBuddy's AI model needed to:

Generate structured 7-day workout plans

Provide concise and actionable nutrition tips

Update workout plans based on user feedback